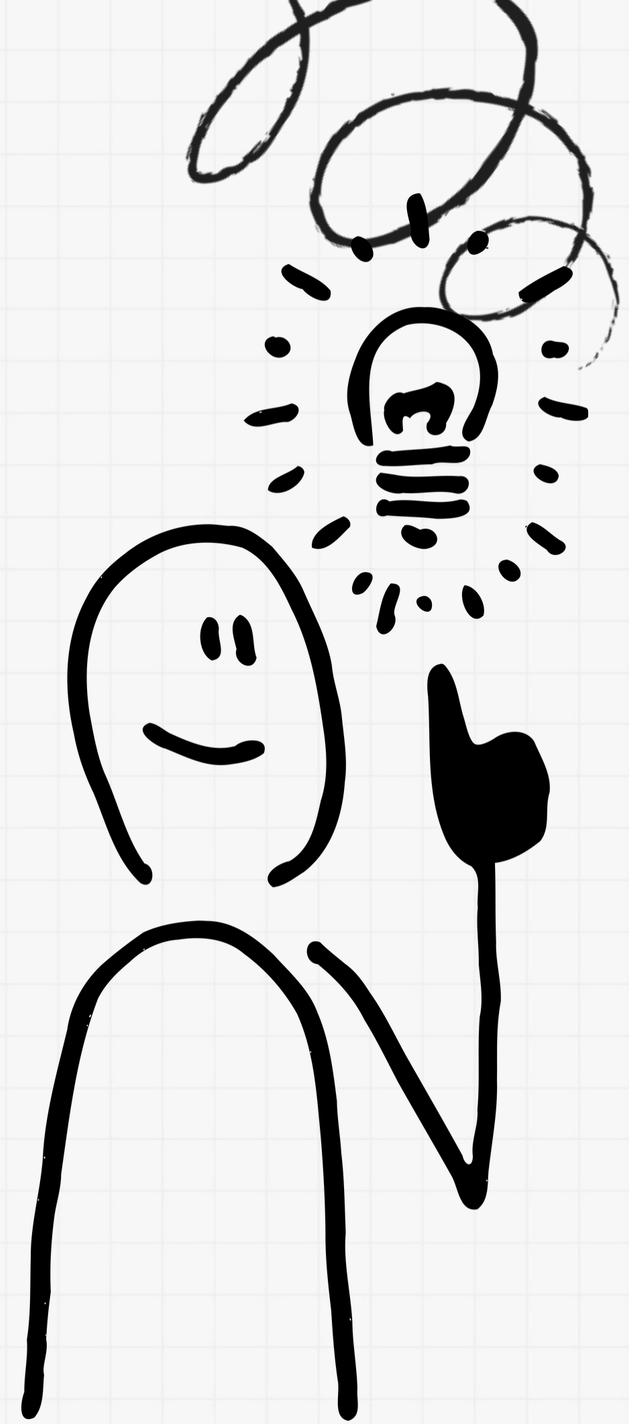
Self-RAG

Learning to Retrieve, Generate, and Critique through Self-Reflection

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University of Washington · Allen Institute for AI · IBM Research AI
ICLR 2024 · arXiv 2310.11511



CS613 · Paper Presentation

Presented by: [Team member names]

LLMs make things up

- **Knowledge lives in the weights.** It is fixed after training, has no source and is hard to update.
- **Rare facts are the worst.** Few training examples, so probability spreads over plausible but wrong answers.
- **Users cannot verify** an answer that comes with no evidence.
- **The standard fix is Retrieval-Augmented Generation (RAG):** look up passages, then answer. But RAG brings problems of its own.



Why not just trust the model?

no retrieval

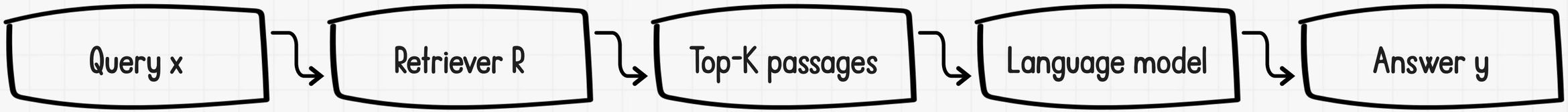
with retrieval

14.7% ⇨ **45.7%**

Llama2-13B accuracy on rare-entity questions (PopQA long-tail: entities with fewer than 100 Wikipedia views a month)



RAG in one slide



dense search over
~21M Wikipedia passages

d1 ... dK
(K = 5 here)

Dense retrieval

$$\mathrm{sim}(q, d) = \langle E(q),\ E(d) \rangle$$

Embed query and passage separately, rank by dot product. Contriever is trained with a contrastive loss; Self-RAG keeps it frozen.

Classic RAG (Lewis 2020)

$$p(y|x) \approx \sum_{z \in \mathrm{top}K} p_\eta(z|x)\, p_\theta(y|x, z)$$

The passage is a latent variable: average the answer over passages, weighted by the retriever. Always retrieves.

In-context RAG (common)

$$y \sim p_\theta(\ \cdot\ |\ [d_1; \ldots; d_K;\ x])$$

Paste all K passages in front of the prompt. No training needed. This is the 'Ret-ChatGPT' baseline.



Why plain RAG falls short



Always retrieves

Top-K passages for every prompt, even 'write an essay about your summer'. Wastes compute and derails creative or reasoning tasks.

Ret-ChatGPT on PubHealth:

70.1 → 54.7



Irrelevant passages

The retriever's top hit can be off-topic, and LMs are easily distracted by irrelevant context (Shi et al. 2023).

Alpaca-13B on PubHealth:

55.5 → 51.1 with retrieval



No grounding check

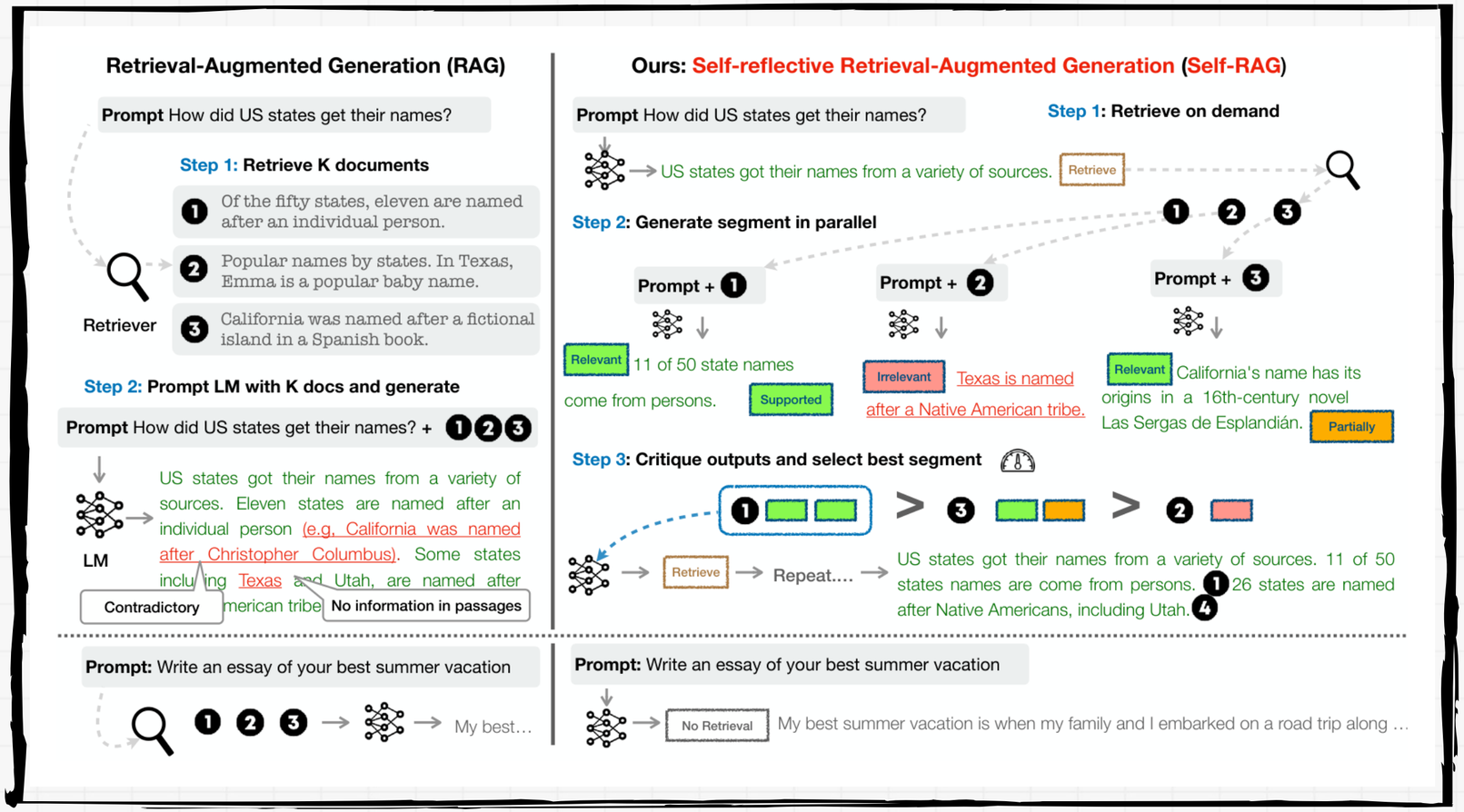
Nothing makes the output follow the passages, and nothing checks it afterwards, so citations are unreliable.

Open RAG models on ASQA:

2–8% citation precision

Earlier fixes each cover one part: FLARE retrieves when confidence is low, SAIL always prepends passages, Toolformer learns API calls, RLHF needs expensive PPO, and self-refine needs many prompt calls.

RAG vs Self-RAG: the paper's picture



Paper Figure 1. Left: standard RAG. Right: Self-RAG on the same prompt. Bottom: a creative prompt.

Plain RAG (left)

Always fetches K = 3 passages and pastes them all into one prompt. The answer mixes them: one claim is contradicted, one has no support.

Self-RAG (right)

- Step 1: retrieve only when needed
- Step 2: one draft per passage, in parallel
- Step 3: critique, keep the best, repeat

Essay prompt (bottom)

Self-RAG writes [No Retrieval] and answers directly. Plain RAG still retrieves.

Core idea: reflection tokens

Add special tokens to the vocabulary and train ONE model, with ordinary next-token prediction, to write them between its own sentences. **Critique becomes next-token prediction.**

Retrieve

Should I fetch evidence for the next sentence?

Yes · No · Continue

sees: x and y<t
before each sentence

IsRel

Is passage d relevant to the question x?

Relevant (best) · Irrelevant

sees: x, d
after the passage

IsSup

Is every claim in my sentence backed by d?

Fully (best) · Partially · No support

sees: x, d, yt
after the sentence

IsUse

Is the whole answer useful? (not about truth)

5 (best) · 4 · 3 · 2 · 1

sees: x, y
once, at the end

What the model actually writes

Q: when did Walking Dead season 7 come out?

[Retrieve=Yes] ...season 7 premiered on October 23, 2016...
[IsRel=Relevant] October 23, 2016 [IsSup=Fully] [IsUse=5]

Q: write a poem about the sunset

[Retrieve=No] The stars begin to twinkle bright... [Retrieve=No] They flicker like a silent dance... [IsUse=5]

Why this order?

- **Retrieve first:** decide before writing
- **IsRel before the sentence:** the judgment shapes the answer
- **IsSup after:** check the finished sentence
- **IsUse last:** a property of the whole answer
- ... passages come from the retriever, not the model

One token stream = chain rule

$$p_{\mathcal{M}}(\tilde{y}|x) = \prod_{t=1}^{T} p(r_t^{ret}) \left[p(r_t^{rel})\, p(y_t)\, p(r_t^{sup}) \right]^{\mathbf{1}[ret_t]} p(y_t)^{1-\mathbf{1}[ret_t]} \cdot p(r^{use})$$

Reflection tokens, formally

Generative story of segment t

$$r_t^{\text{ret}} \sim p_{\mathcal{M}}(\,\cdot \mid x,\, y_{<t})$$ decide: retrieve?

$$d_t = \mathcal{R}(x,\, y_{t-1}) \quad \text{only if} \quad r_t^{\text{ret}} = \text{Yes}$$ external call, not modelled

$$r_t^{\text{rel}} \sim p_{\mathcal{M}}(\,\cdot \mid x,\, d_t,\, y_{<t})$$ judge the passage

$$y_t \sim p_{\mathcal{M}}(\,\cdot \mid x,\, d_t,\, r_t^{\text{rel}},\, y_{<t})$$ write the sentence

$$r_t^{\text{sup}} \sim p_{\mathcal{M}}(\,\cdot \mid x,\, d_t,\, y_t,\, \ldots)$$ check the sentence

$$r^{\text{use}} \sim p_{\mathcal{M}}(\,\cdot \mid x,\, y)$$ after the last segment

One token stream = chain rule

$$p_{\mathcal{M}}(\tilde{y} \mid x) = \prod_{t=1}^{T} p(r_t^{\text{ret}}) \left[p(r_t^{\text{rel}})\, p(y_t)\, p(r_t^{\text{sup}})\right]^{\mathbb{1}_t} p(y_t)^{1-\mathbb{1}_t} \cdot\; p(r^{\text{use}})$$

Passages are inputs from the retriever R, not outputs of M, so they are not in the product.

Reading a decision from logits

$$p(r \mid r \in G) = \frac{e^{z_r}}{\sum_{r' \in G} e^{z_{r'}}} = \frac{p(r)}{\sum_{r' \in G} p(r')}$$

Each token type G is a small group (Yes / No, Relevant / Irrelevant, ...). Renormalising inside the group removes the ~32,000 ordinary tokens: the global softmax denominator cancels. This is how the threshold δ and the critique scores are computed.

New vocabulary

$$\mathcal{V}' = \mathcal{V} \cup \mathcal{R}_{\text{tokens}}$$

New rows in the embedding table and output layer, trained like any other word. A segment = one sentence (split with spaCy).

Training: distil a critic, then teach the generator

GPT-4 labels → **Critic C** → **Annotate 150k** → **Generator M**

Few-shot prompts per token type: 12.6k Retrieve, 19.3k IsRel, 11.2k IsSup, 3.8k IsUse

Llama2-7B fine-tuned on those labels. Agrees with GPT-4: 94% Retrieve, 94% IsSup, 80% IsRel, 74% IsUse

For each sentence: C decides retrieval, R fetches, C labels IsRel / IsSup, keeps one passage. Text unchanged

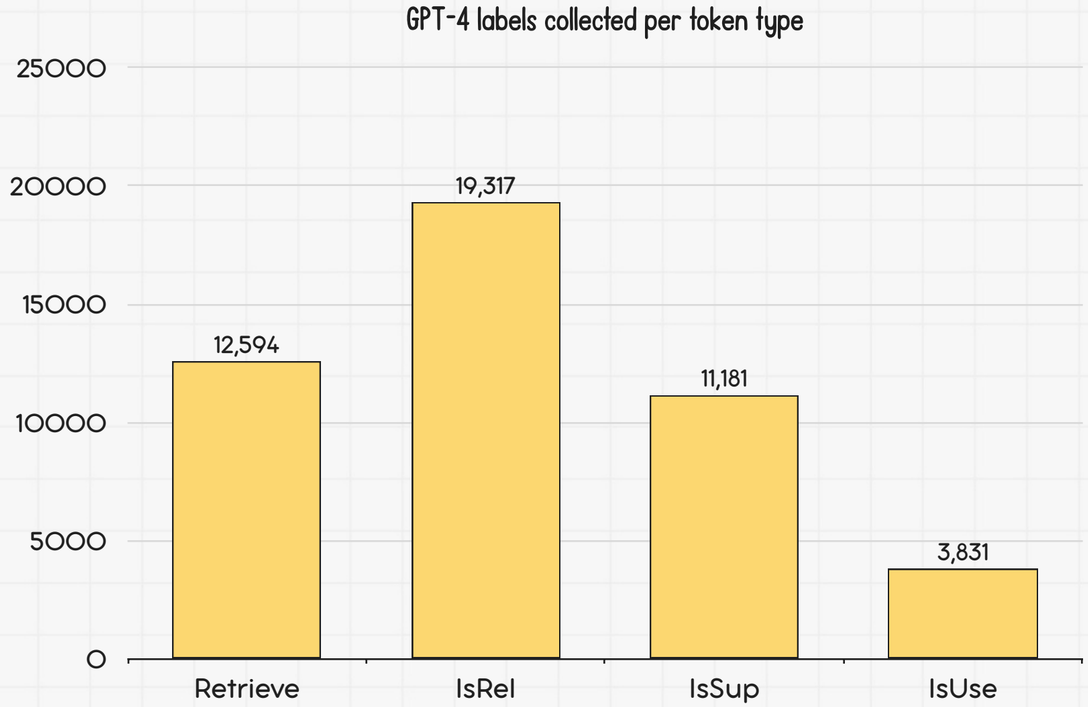
Llama2 7B / 13B + new tokens, plain next-token training (3 epochs, 4×A100)

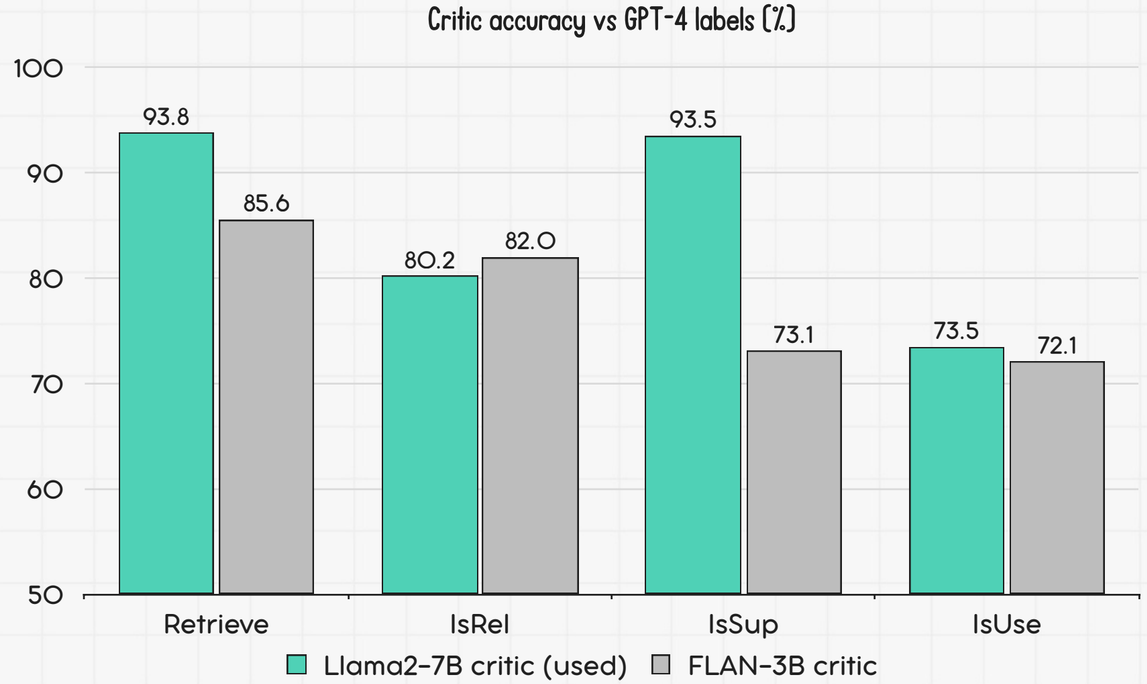
At inference: only M + the retriever

C and GPT-4 are only used to build the training data. M has learned to be its own critic.



Stage 1: GPT-4 teaches the critic





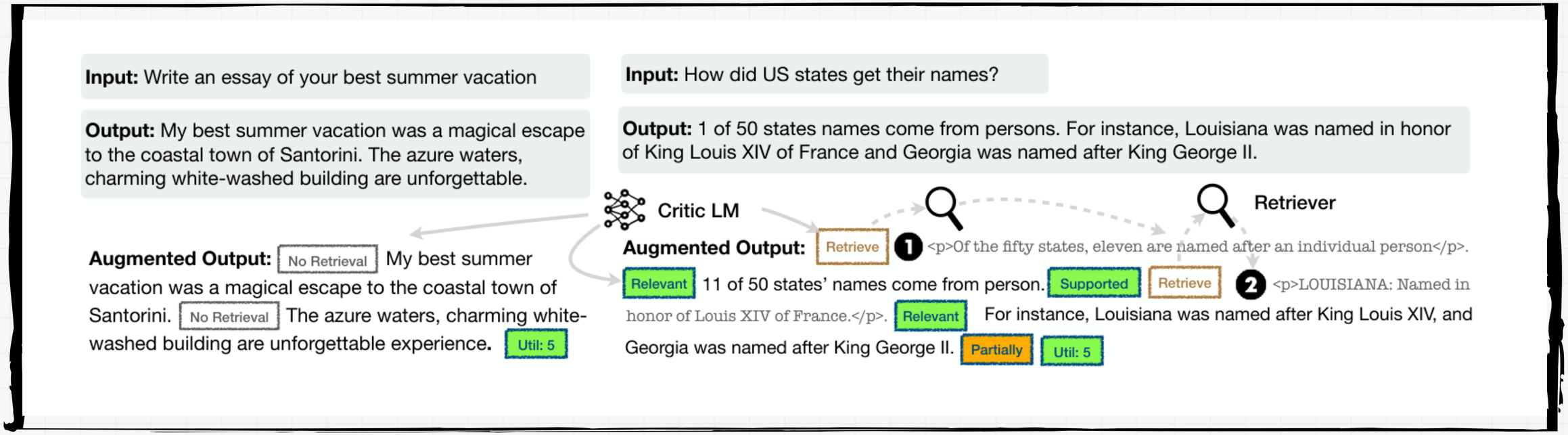
Is GPT-4 itself right?

Authors hand-checked 20 labels per type: agreement 95% Retrieve, 95% IsRel, 90% IsSup, 80% IsUse (errors mostly 4 vs 5 or 1 vs 2).

Settings

One instruction + few-shot demos per token type (App. D) · temperature 1 · max 200 tokens · wrong formats dropped · about 47k labels in total · critic trained with Eq. 1

Stage 2: the critic labels 145k examples



Paper Figure 2: two training targets. Left: no retrieval needed. Right: the critic inserts Retrieve, passages, IsRel, IsSup and IsUse.

STEP 01

C predicts Retrieve for (x, y). No: add only IsUse, done.

STEP 02

Yes: retrieve with (x, y) as the query; split y into sentences (spaCy).

STEP 03

Per sentence: C says Yes / No / Continue. Yes: retrieve with (x, that sentence).

STEP 04

Per passage: C labels IsRel and IsSup. Keep the best relevant + supported one; if none, a random one.

STEP 05

C adds IsUse at the end. The answer text itself is never rewritten.

The two training objectives

Critic · Eq. 1

$$\max_{C} \; \mathbb{E}_{((x,y),r)\sim\mathcal{D}_{critic}} \log p_C(r \mid x, y)$$

A classifier written as language modelling: predict GPT-4's label r as text.

Generator · Eq. 2

$$\max_{\mathcal{M}} \; \mathbb{E}_{(x,y,r)\sim\mathcal{D}_{gen}} \log p_{\mathcal{M}}(y, r \mid x)$$

$$\mathcal{L} = -\sum_i m_i \log p_{\mathcal{M}}(\tilde{w}_i \mid x, \tilde{w}_{<i}), \quad m_i = 0 \text{ inside } \langle p\rangle \ldots \langle /p\rangle$$

Writes the answer AND its critique, interleaved.

Why mask the passages?

- They are inputs from the retriever, not model output
- Predicting them would teach memorising Wikipedia, or making up passages
- The model still attends to them, so it learns to use them

Why not RLHF / PPO?

- PPO: about 4 models in memory and on-policy sampling
- Self-RAG: 1 model, static corpus, cross-entropy loss
- Behaviour is changed later at inference, without retraining

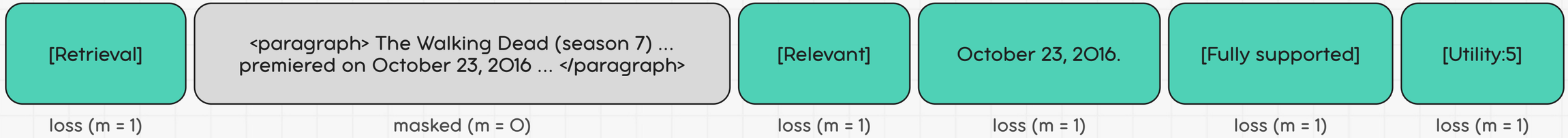
The generator loss, token by token

$$\mathcal{L}_{\text{gen}} = -\sum_{i=1}^{n} m_i \log p_{\mathcal{M}}(\tilde{w}_i \mid x,\ \tilde{w}_{<i})$$

$$m_i = 0 \ \text{ inside } \langle p\rangle \ldots \langle /p\rangle, \qquad m_i = 1 \ \text{ otherwise}$$

Ordinary next-token cross-entropy. The only change: tokens inside a retrieved passage get no loss.

One training sequence



What changes in the model

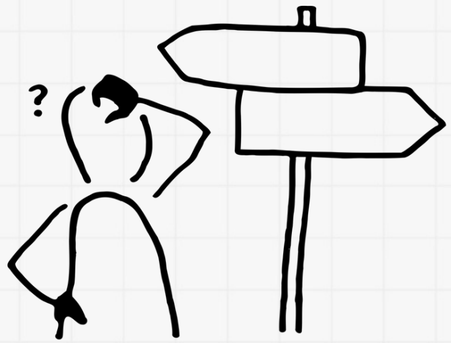
The vocabulary grows by the reflection tokens: new rows in the embedding and output matrices. The model still reads passages through attention; it just never learns to write them.

Training recipe (App. A.2)

- Llama2 7B / 13B, 3 epochs, batch size 128
- Peak LR 2e-5, 3% warm-up, linear decay
- Max length 2,048 (7B) / 1,524 (13B)
- 4x A100 80GB, DeepSpeed ZeRO-3, bf16, FlashAttention

Inference, one sentence at a time

$$\text{retrieve} \Leftrightarrow \frac{p(\text{Yes})}{p(\text{Yes}) + p(\text{No})} > \delta$$



Probability of 'Yes' within the Yes/No group. Default δ = O.2: retrieve even when 'No' is more likely. δ = O for ASQA, where citations are required.

STAGE 01 → STAGE 02 → STAGE 03 → STAGE 04 → STAGE 05

Retrieve?

Check the threshold above. If no: write the sentence like a normal LM

Fetch K

R returns the top 5 passages for (x, previous sentence)

K drafts

One candidate sentence per passage, in parallel, each after its IsRel token

Critique

IsSup for each (d, sentence) pair; IsUse for usefulness

Keep best B

Segment-level beam search, B = 2, ranked by the score f

Algorithm 1, line by line

Algorithm 1 SELF-RAG Inference

Require: Generator LM $\mathcal{M}$, Retriever $\mathcal{R}$, Large-scale passage collections $\{d_1, \ldots, d_N\}$

1: **Input:** input prompt x and preceding generation $y_{<t}$, **Output:** next output segment y_t
2: $\mathcal{M}$ predicts Retrieve given $(x, y_{<t})$
3: **if** Retrieve == Yes **then**
4: Retrieve relevant text passages $\mathbf{D}$ using $\mathcal{R}$ given (x, y_{t-1}) ▷ Retrieve
5: $\mathcal{M}$ predicts IsRel given x, d and y_t given $x, d, y_{<t}$ for each $d \in \mathbf{D}$ ▷ Generate
6: $\mathcal{M}$ predicts IsSup and IsUse given x, y_t, d for each $d \in \mathbf{D}$ ▷ Critique
7: Rank y_t based on IsRel , IsSup , IsUse ▷ Detailed in Section 3.3
8: **else if** Retrieve == No **then**
9: $\mathcal{M}_{gen}$ predicts y_t given x ▷ Generate
10: $\mathcal{M}_{gen}$ predicts IsUse given x, y_t ▷ Critique

Lines 2–3: decide

Retrieve is read as P(Yes) / (P(Yes) + P(No)) > δ, not argmax. Default δ = 0.2.

Line 4: retrieve

Query = (x, previous segment). Top-5 passages from Contriever-MS MARCO.

Lines 5–7: draft, judge, rank

One draft per passage, in parallel. IsRel before the draft; IsSup and IsUse after. Rank by the score f.

Lines 8–10: no retrieval

Works like a normal LM; still predicts IsUse.

Cost and special cases

- Up to B × K = 2 × 5 = 10 segment drafts per retrieval step, run in parallel with vLLM
- Greedy decoding inside a segment, beam search over segments
- PubHealth / ARC: scores are summed per answer option; best option wins
- The paper reports no wall-clock latency

Scoring a candidate sentence

$$f(y_t, d) = p(y_t \mid x, d, y_{<t}) + w^{Rel} s^{Rel} + w^{Sup} s^{Sup} + w^{Use} s^{Use}$$

$$s^{Rel} = \frac{p(\text{Relevant})}{p(\text{Rel}) + p(\text{Irrel})}$$

$$s^{Sup} = \frac{p(\text{Fully}) + 0.5\, p(\text{Partially})}{p(\text{Fully}) + p(\text{Partially}) + p(\text{No})}$$

$$s^{Use} = \sum_{i=1}^{5} u_i \frac{p(i)}{\sum_j p(j)}, \quad u = (-1, -\tfrac{1}{2}, 0, \tfrac{1}{2}, 1)$$

- **Defaults:** w = (1, 1, 0.5)
- **Expected value** of each critique, in a fixed range
- **Hard mode:** drop any candidate with IsSup = No support

Worked example

(illustrative numbers)

	d1: states named after people	d2: baby names in Texas
Fluency p(yt)	0.60	0.70
s Rel	0.90	0.20
s Sup	0.80	0.25
s Use	0.57	0.33
Score f	2.59 (chosen)	1.31

Likelihood alone would pick d2, a fluent sentence built on an off-topic passage. The critique terms pick the grounded d1.

Why it works

Decomposition

4 small, separately supervised decisions instead of one hard one

Retrieve if useful

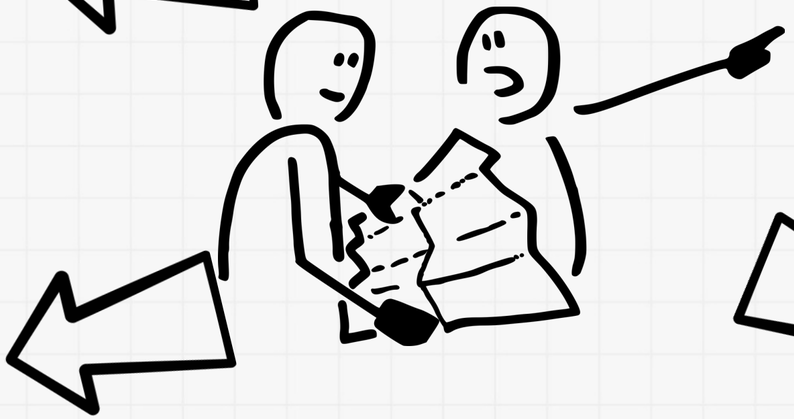
Essays and maths stay a normal LM, so versatility is kept

Generate + verify

Checking a sentence against one passage is easier than writing a correct one

Cheap critique

Learned offline with SFT: no RL, no reward model at test time



One passage each

No mixing of noisy passages: bad ones are scored down

RLHF in disguise

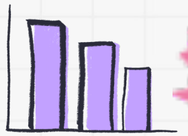
Like $\pi^* = \pi_{ref} \cdot \exp(\text{reward} / \beta) / Z$: changing w changes the reward, no retraining

Experimental setup



Closed-set

- PubHealth: health-claim fact check
- ARC-Challenge: science multiple choice
- Metric: accuracy



Short-form QA

- PopQA long-tail: 1,399 rare-entity questions
- TriviaQA: 11,313 questions
- Metric: gold answer appears in the output



Long-form

- Biographies: FactScore (% of atomic facts supported)
- ASQA: correctness, fluency (MAUVE), citation precision and recall

Baselines No retrieval: Llama2, Alpaca, Llama2-chat, ChatGPT, CoVE-65B · Standard RAG: the same models + the same passages · Same data, no reflection: Llama2-FT · Retrieval-trained: SAIL, Toolformer

Settings Llama2 7B / 13B · 150k training pairs · Contriever-MS MARCO top-5 (+5 web results for Bio and QA) · zero-shot · δ = 0.2, B = 2, w = (1, 1, 0.5)

What each metric measures

PopQA, TriviaQA: match accuracy

$$\mathrm{Acc} = \frac{1}{N}\sum_{j=1}^{N} \mathbb{1}\left[\exists\, a \in \mathcal{A}_j :\ a \subseteq \hat{y}_j\right]$$

Correct if any gold alias appears in the output. Caveat: a long answer can match by luck.

Bio: FactScore

$$\mathrm{FS}(y) = \frac{1}{|\mathcal{A}_y|}\sum_{a \in \mathcal{A}_y} \mathbb{1}\left[a \text{ supported by Wikipedia}\right]$$

Split the bio into atomic facts; check each against Wikipedia. Precision, not coverage: short bios score well.

ASQA: citation recall, precision

$$\mathrm{rec}_i = \mathbb{1}\left[\mathrm{concat}(C_i) \models s_i\right]$$

$$c \text{ irrelevant} \Leftrightarrow \neg(c \models s_i) \quad \wedge \quad C_i \setminus \{c\} \models s_i$$

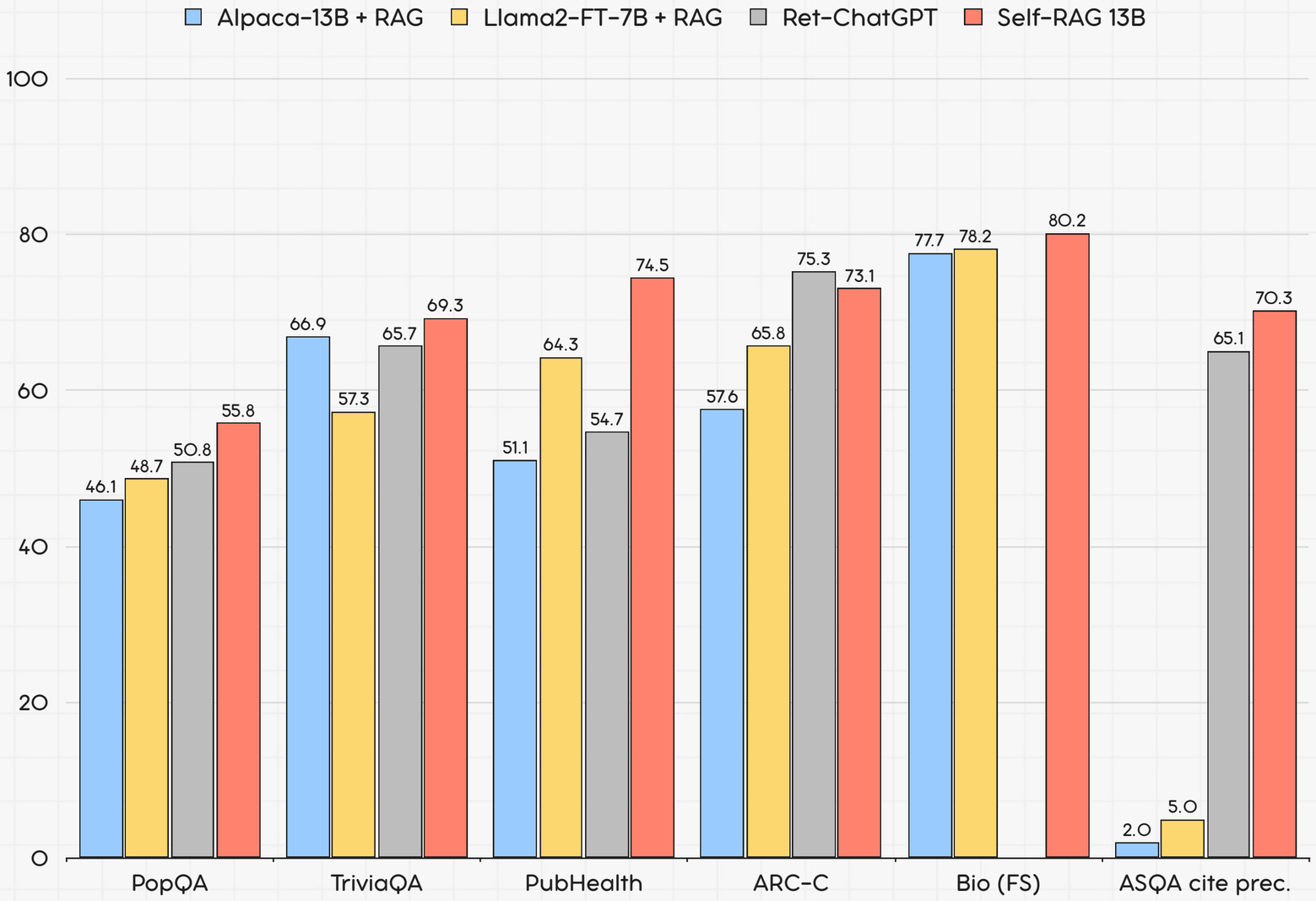
Judged by an NLI model (TRUE, T5-11B). Recall: is the claim backed? Precision: is every citation needed?

ASQA: fluency and correctness

$$\mathrm{MAUVE} = \mathrm{AUC}\left\{\left(e^{-c\,\mathrm{KL}(Q\|R_\lambda)},\ e^{-c\,\mathrm{KL}(P\|R_\lambda)}\right)\right\}, \quad R_\lambda = \lambda P + (1-\lambda)Q$$

MAUVE: how close model text is to human text (1 = same). str-em: share of short gold answers found in the output.

Main results



- **Best open model** on all six tasks
- **Beats ChatGPT** on PopQA, PubHealth, Bio and citation precision
- **Not everywhere:** ChatGPT without retrieval still wins TriviaQA (74.3) and ARC (75.3)
- **Same data is not enough:** Llama2-FT trails, so the gain comes from the framework

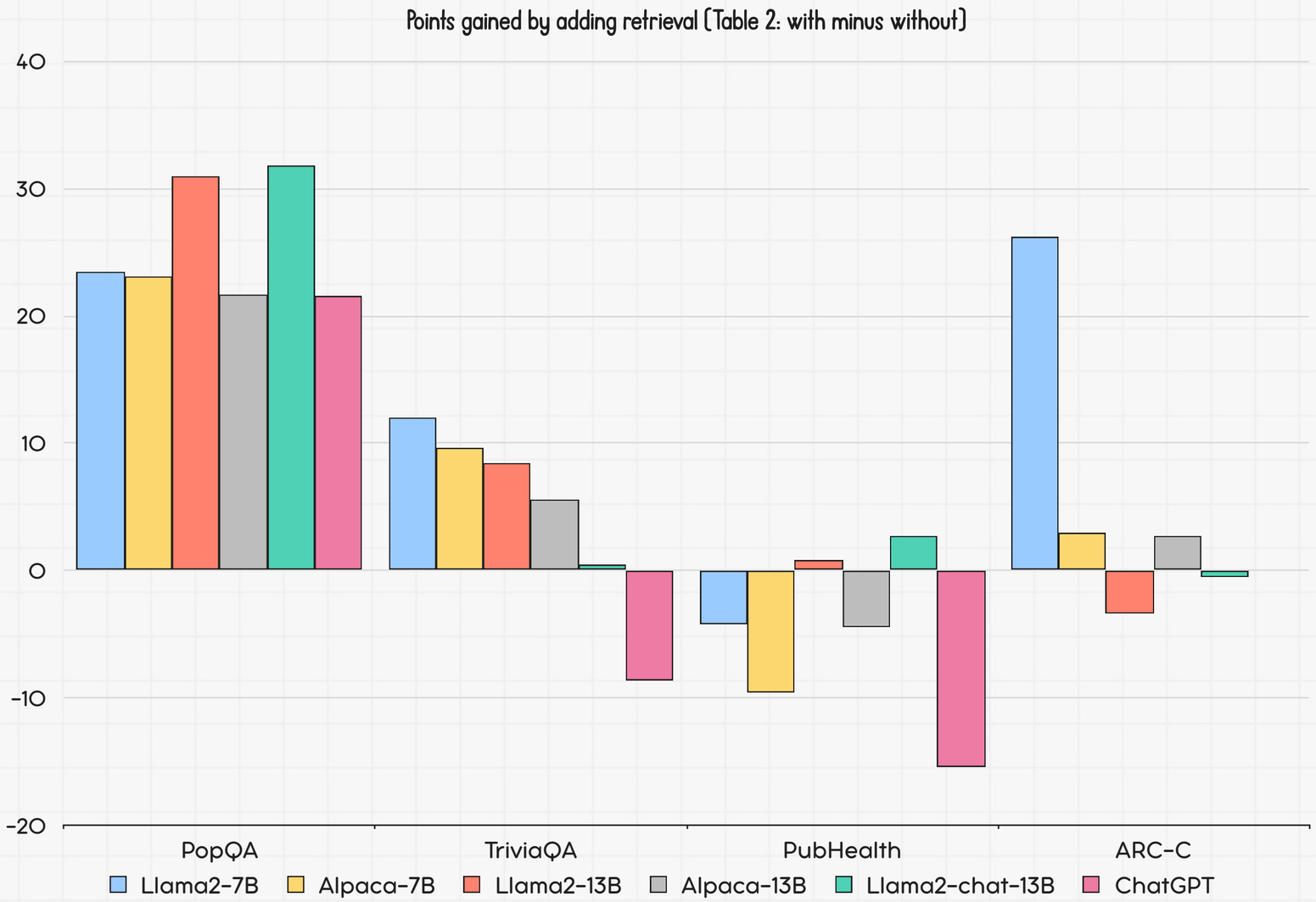
Source: Table 2. Ret-ChatGPT has no Bio score.

Table 2 in full

Model	Size	PopQA	TQA	PubHealth	ARC-C	Bio FS	ASQA em	rouge	MAUVE	prec	rec
LMs with proprietary data											
Llama2-chat	13B	20.0	59.3	49.4	38.4	55.9	22.4	29.6	28.6	-	-
Ret-Llama2-chat	13B	51.8	59.8	52.1	37.9	79.9	32.8	34.8	43.8	19.8	36.1
ChatGPT		29.3	74.3	70.1	75.3	71.8	35.3	36.2	68.8	-	-
Ret-ChatGPT		50.8	65.7	54.7	75.3	-	40.7	39.9	79.7	65.1	76.6
Perplexity.ai		-	-	-	-	71.2	-	-	-	-	-
Baselines without retrieval											
Llama2	7B	14.7	30.5	34.2	21.8	44.5	7.9	15.3	19.0	-	-
Alpaca	7B	23.6	54.5	49.8	45.0	45.8	18.8	29.4	61.7	-	-
Llama2	13B	14.7	38.5	29.4	29.4	53.4	7.2	12.4	16.0	-	-
Alpaca	13B	24.4	61.3	55.5	54.9	50.2	22.9	32.0	70.6	-	-
CoVE*	65B	-	-	-	-	71.2	-	-	-	-	-
Baselines with retrieval											
Toolformer*	6B	-	48.8	-	-	-	-	-	-	-	-
Llama2	7B	38.2	42.5	30.0	48.0	78.0	15.2	22.1	32.0	2.9	4.0
Alpaca	7B	46.7	64.1	40.2	48.0	76.6	30.9	33.3	57.9	5.5	7.2
Llama2-FT	7B	48.7	57.3	64.3	65.8	78.2	31.0	35.8	51.2	5.0	7.5
SAIL*	7B	-	-	69.2	48.4	-	-	-	-	-	-
Llama2	13B	45.7	47.0	30.2	26.0	77.5	16.3	20.5	24.7	2.3	3.6
Alpaca	13B	46.1	66.9	51.1	57.6	77.7	34.8	36.7	56.6	2.0	3.8
Ours											
Self-RAG	7B	54.9	66.4	72.4	67.3	81.2	30.0	35.7	74.3	66.9	67.8
Self-RAG	13B	55.8	69.3	74.5	73.1	80.2	31.7	37.0	71.6	70.3	71.3

Yellow = best non-proprietary score per column. Green = Self-RAG. * = number copied from the original paper.

Retrieval helps some tasks, hurts others



+31.8

Llama2-chat-13B on PopQA: rare facts can be copied from a passage

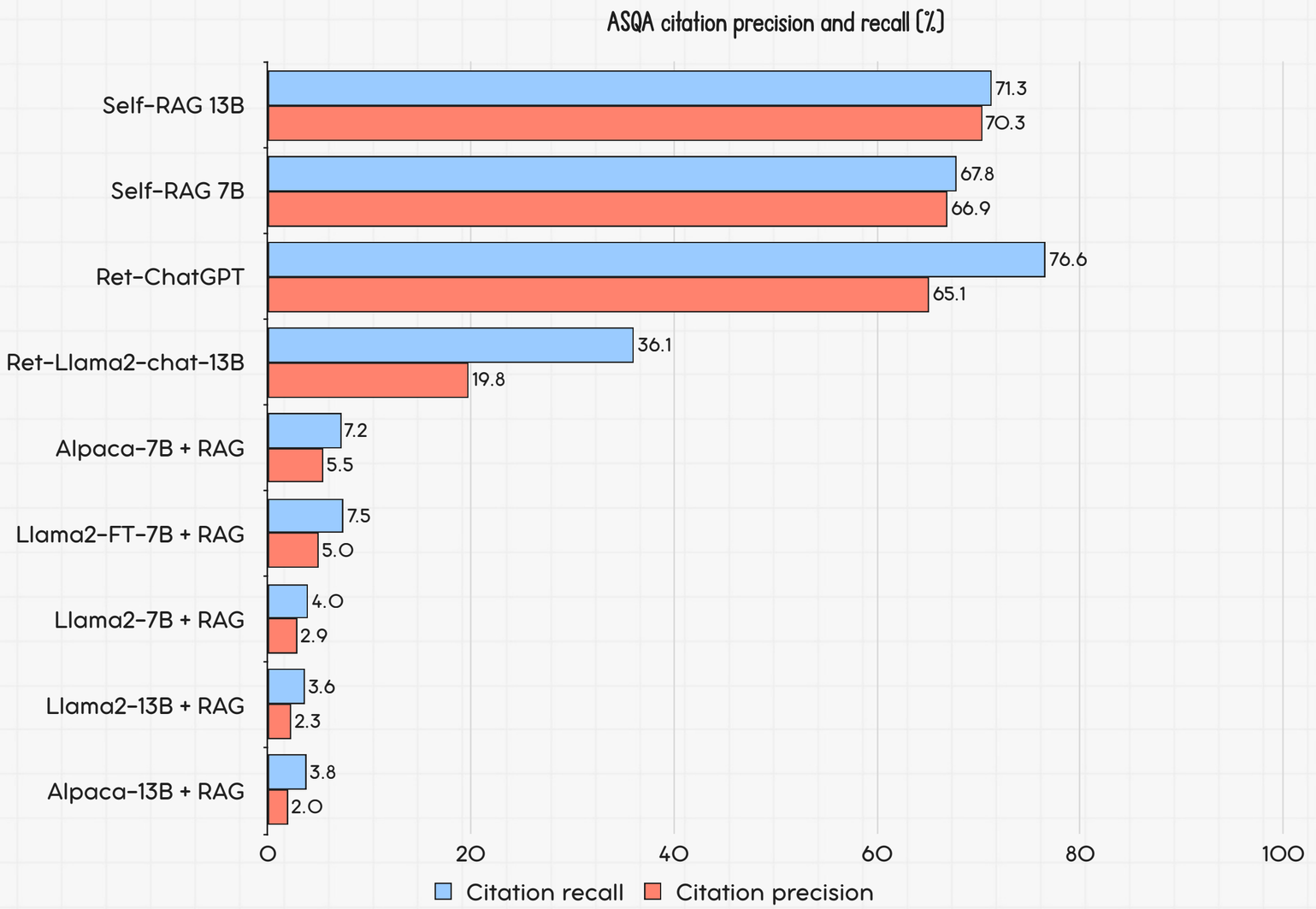
-15.4

ChatGPT on PubHealth: claim checking needs reasoning, and passages distract

So: decide per query

That is the Retrieve token. Self-RAG never retrieving: PopQA 45.5 → 24.7, but PubHealth 73.5 → 73.0 (Fig. 3a).

Citations and grounding



Answers from memory

Of the correct PopQA / TriviaQA answers, how many are NOT in the retrieved passages? (App. C.1)

2%

Self-RAG

15-20%

Alpaca, Llama2-chat

Short but precise

Bio FactScore: 7B (81.2) beats 13B (80.2) because it writes shorter, fully grounded bios. With an irrelevant passage, Self-RAG says IsRel = Irrelevant instead of bluffing.

Ablations: every piece matters

7B, 50k training examples	PopQA	PubHealth	ASQA em
Self-RAG (full)	45.5	73.5	32.1
Train: no retriever	43.6	67.8	31.0
Train: no critic	42.6	72.0	18.1
Test: never retrieve	24.7	73.0	-
Test: hard threshold	28.3	72.6	-
Test: always top-1 passage	41.8	73.1	28.6
Test: drop IsSup score	44.1	73.2	30.6

- **No critic → ASQA collapses:** reflection tokens are the core
- **No retrieval hurts PopQA, not PubHealth:** needs differ by task
- **Hard threshold retrieves too rarely:** δ is a crucial knob

Test-time knobs (no retraining)

w Sup 0 → 2

citation precision 69.7 → 70.5, but MAUVE fluency 96 → 86

δ up

fewer retrievals; costs PopQA much more than PubHealth

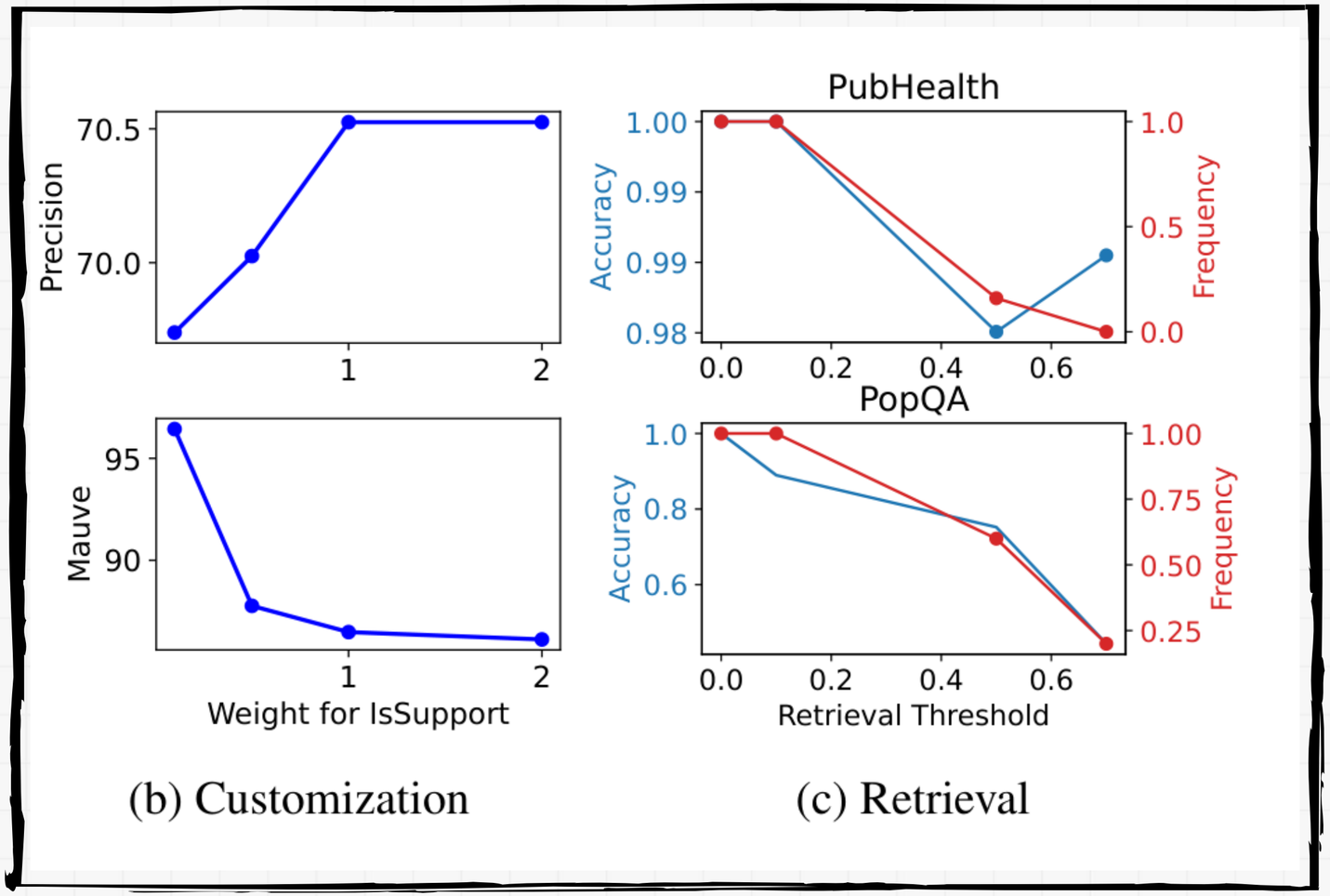
5k → 150k data

PopQA about 34 → 55 and still rising

Human check

IsRel judged correct 90–95%, IsSup 85–90%

Test-time knobs (Figure 3b, 3c)



Paper Figure 3 (b) and (c). In (c), accuracy is normalised to the best setting.

3b: weight on IsSup

w Sup O → 2: citation precision about 69.7 → 70.5, MAUVE fluency about 96 → 86.

More weight on support = safer citations but less fluent text. You pick the point at inference.

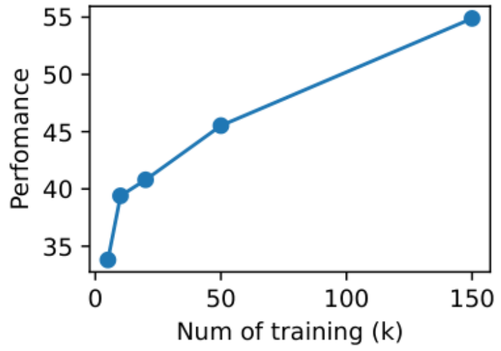
3c: retrieval threshold δ

Higher δ = fewer retrievals.

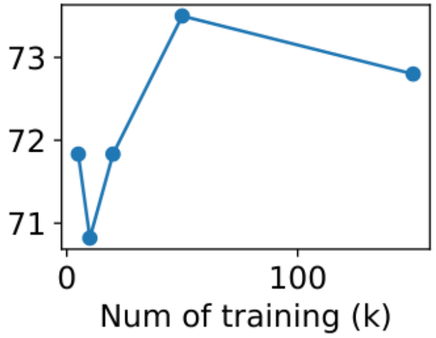
PubHealth: accuracy barely moves even at 0% retrieval. PopQA: accuracy falls with retrieval frequency.

So cost can be traded for accuracy per task.

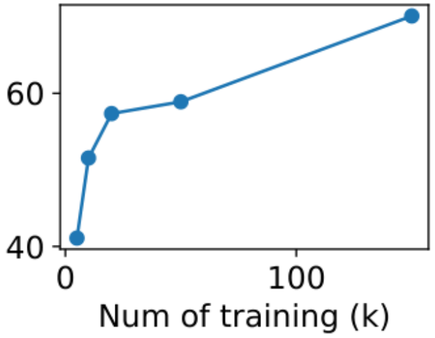
More data, and a human check (Figure 4)



(a) PopQA



(b) PubHealth



(c) ASQA (prec)

	Pop	Bio.
S & P	92.5	70.0
ISREL	95.0	90.0
ISSUP	90.0	85.0

(d) Human evaluation on PopQA and Bio generation.

(a) PopQA

About 34 → 55 from 5k to 150k examples, still rising.

(b) PubHealth

Flat, about 71 to 73.5: this task needs little retrieval skill.

(c) ASQA precision

About 41 → 70, still rising. More data would likely help.

(d) Human check

50 samples each. Supported and plausible: 92.5% PopQA, 70% Bio. IsRel right 95 / 90%, IsSup 90 / 85%.

The paper notes Llama2-FT shows no comparable gain from 50k to 150k examples: more data helps because of the reflection tokens.

Critical analysis

Strengths

- No RL: SFT + reranking
- Tunable at test time (δ, w)
- Per-sentence citations with self-labels
- Strong controls: Llama2-FT, ablations

Limitations

- Every label traces back to GPT-4
- Critic errors become training targets
- Frozen retriever: detects bad passages, cannot fix them
- Up to $B \cdot K = 10$ drafts per step; latency not reported

Assumptions

- One sentence ← one passage
- GPT-4 judgments match human ones
- The corpus contains the facts
- Token probabilities can simply be added

Improvements

- Re-query when every IsRel is low
- Learn or calibrate w on a dev set
- Let a sentence use several passages
- Preference training (DPO) on critique pairs

Impact

2020

RAG

Lewis et al.: retrieval as a latent variable, always on

2023

Self-RAG

Retrieval on demand + self-critique tokens (ICLR 2024)

2024

CRAG, Adaptive-RAG

Grade documents, fall back to web search; route queries by difficulty

2024

Agentic RAG

LangGraph / LlamaIndex recipes: retrieval, hallucination and answer 'graders' = IsRel / IsSup / IsUse

2025

Search as a tool

Assistants decide when to search and cite; RL-trained search-and-reason models

Lasting idea: special tokens as an interface for control and self-evaluation, and test-time reranking instead of retraining.



Team Contribution



Student	Contribution	Score	Main work
[Student A]	[Equal / Partial]	[1.0]	[e.g. problem & motivation, slides 2–5]
[Student B]	[Equal / Partial]	[1.0]	[e.g. method & equations, slides 6–17]
[Student C]	[Equal / Partial]	[1.0]	[e.g. experiments, slides 18–26]
[Student D]	[Equal / Partial]	[1.0]	[e.g. critique & impact, slides 27–28]

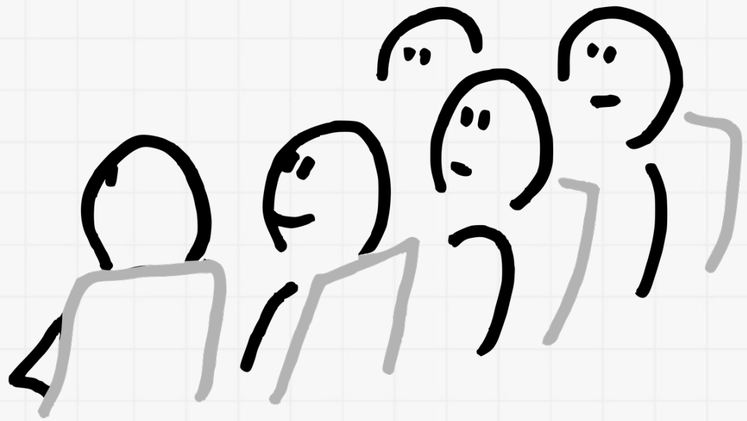
Scores reflect actual relative contribution (preparation, understanding, slides, presentation), as the rubric requires.

Thank you!

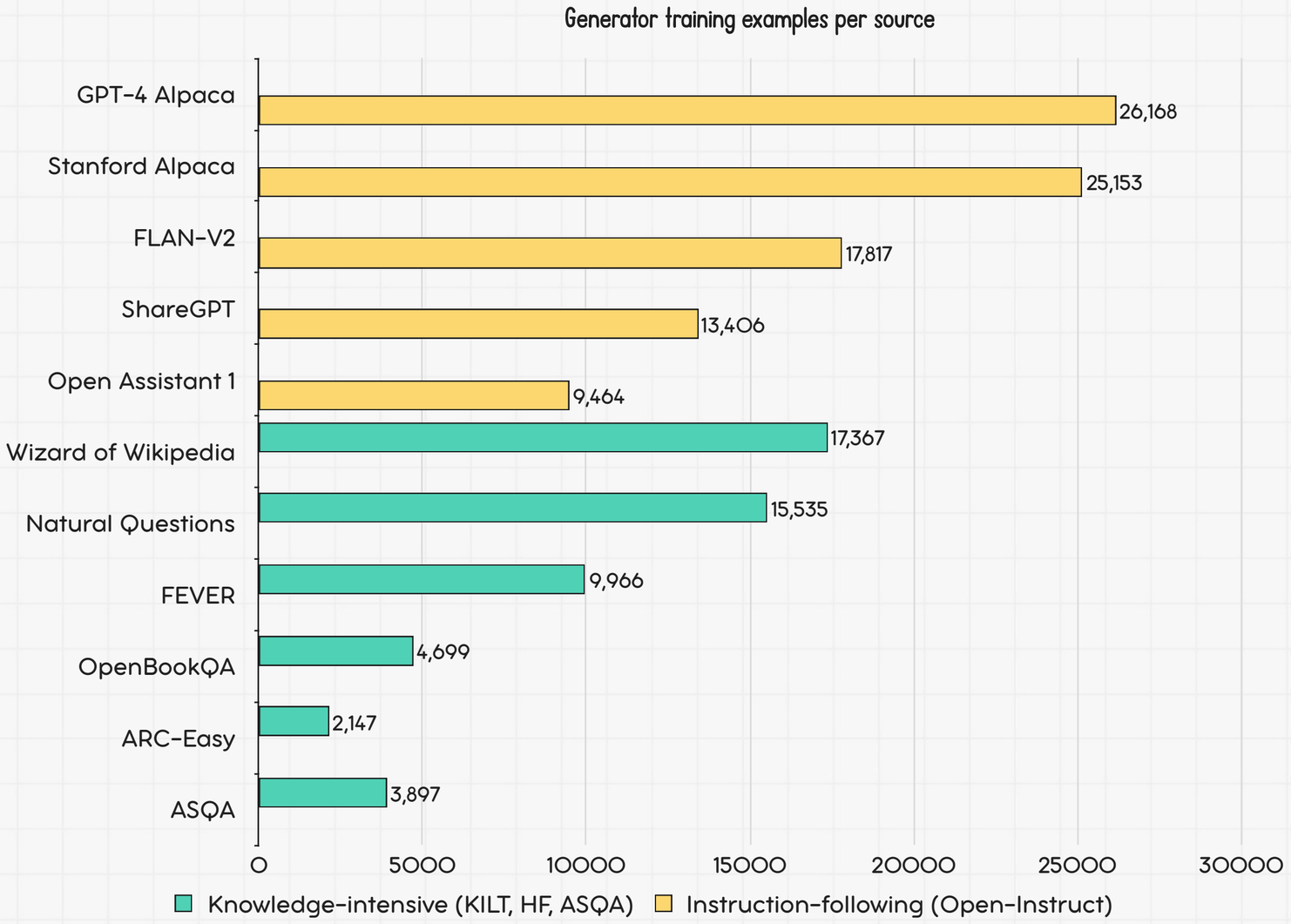
Questions?

Paper: arxiv.org/abs/2310.11511
Code and models: selfrag.github.io





Backup: the training data (Table 3)



145,619

examples in total (the paper rounds to 150k)

- 92,008 instruction-following
- 53,611 knowledge-intensive
- Same pairs train Llama2-FT, minus tokens and passages

Fair Q&A point

ARC-Easy and ASQA training data are related to the ARC-Challenge and ASQA test sets, so these two tasks are not fully out-of-domain.

Backup: settings to know for Q&A

Setting	Value (paper §4.2, App. A.2, B.2)
Retriever	Contriever-MS MARCO, frozen. Top-5 Wikipedia passages (2018 dump; Dec 2020 for PopQA)
Extra evidence	+5 Google Programmable Search results for Bio and open QA; ASQA: GTR-XXL top-5 for every system
Retrieval threshold	δ = 0.2 by default; δ = 0 for ASQA (citations required)
Critique weights	w Rel = 1.0, w Sup = 1.0, w Use = 0.5
Decoding	Beam of 2 over segments, greedy inside a segment, vLLM
Max new tokens	100 open QA, 50 closed-set, 200 per segment for Bio, 100 for ASQA baselines
Critic	Llama2-7B; GPT-4 labels at temperature 1, max 200 tokens
Training data creation	Up to 10 retrieved documents per input
Hardware	Training 4× A100 80GB; inference 1–2 Quadro RTX 6000 24GB